

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Data Analyst

Python | PostgreSQL | SQL | Power BI

01. Business Problem + Approach

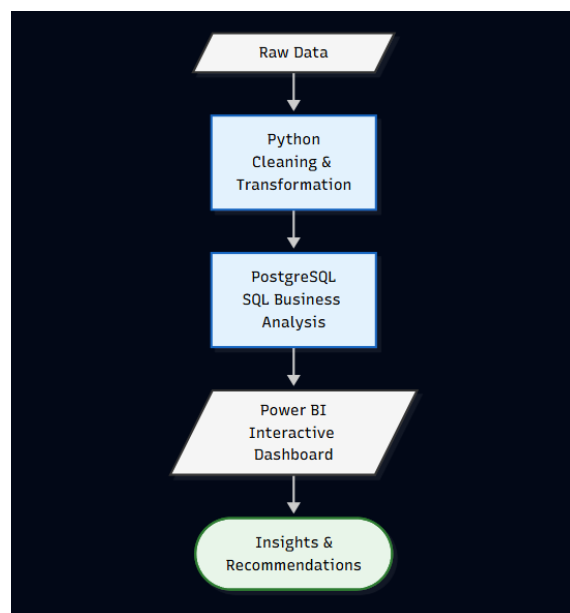
Business Problem

A retail company wants to understand the customer shopping behavior to improve customer engagement, retention, marketing, and product strategy.

Business Question

How can customer shopping data be used to identify trends, improve customer engagement, and optimize marketing and product strategies?

Approach



Dataset

3,900 record and 18 columns

Includes:

- Customer demographics
- Purchase information
- Product categories
- Discounts
- Reviews
- Shipping
- Subscription status
- Previous purchases

02. Data Preparation + SQL

Data Preparation

- Inspect structure and data quality with **Pandas**
- Identified **37** missing Review Rating values
- Imputed missing ratings using category-level median ratings
- Standardized column names to **snake_case**
- Created **age_group** and **purchase_frequency_days**
- Remove redundant **promo_code_used**
- Loaded cleaned data into **PostgreSQL**

SQL Analysis

It was used to answer business questions covering:

- **Revenue:** gender, category, and age group
- **Customers:** subscription and loyalty
- **Products:** ratings and category rankings
- **Behavior:** discounts, shipping, and repeat purchases

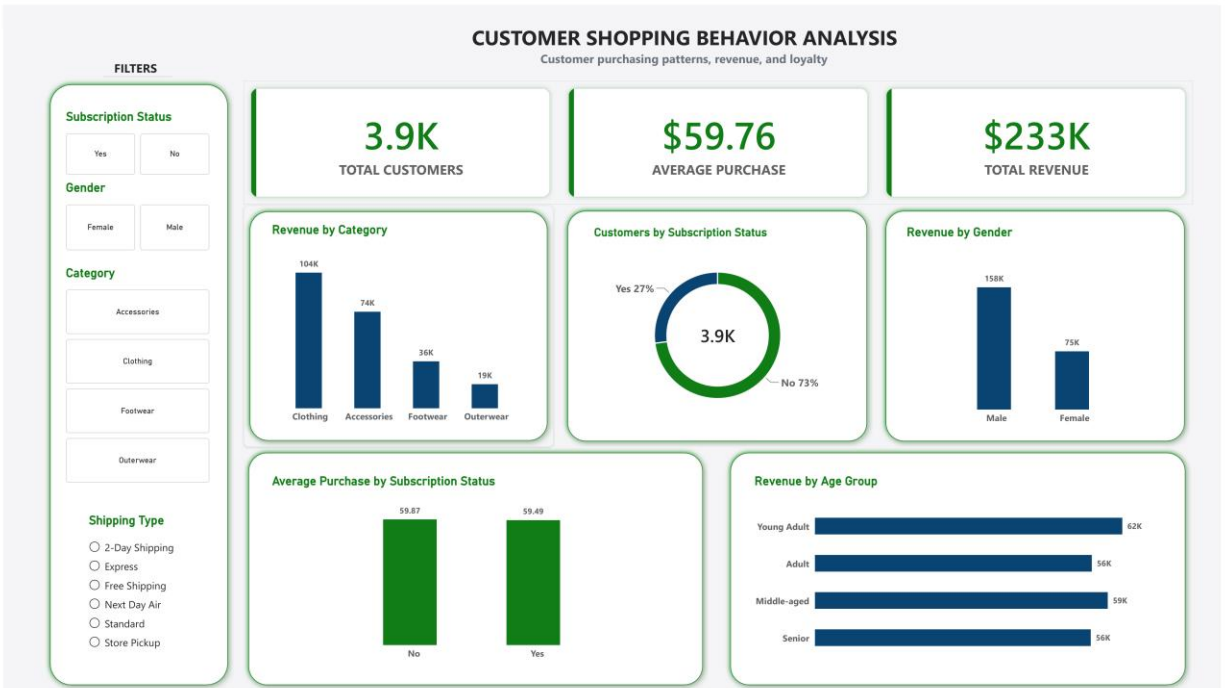
Customer Segmentation

Segment	Previous Purchases
New	1
Returning	2-10
Loyal	>10

03. Power BI

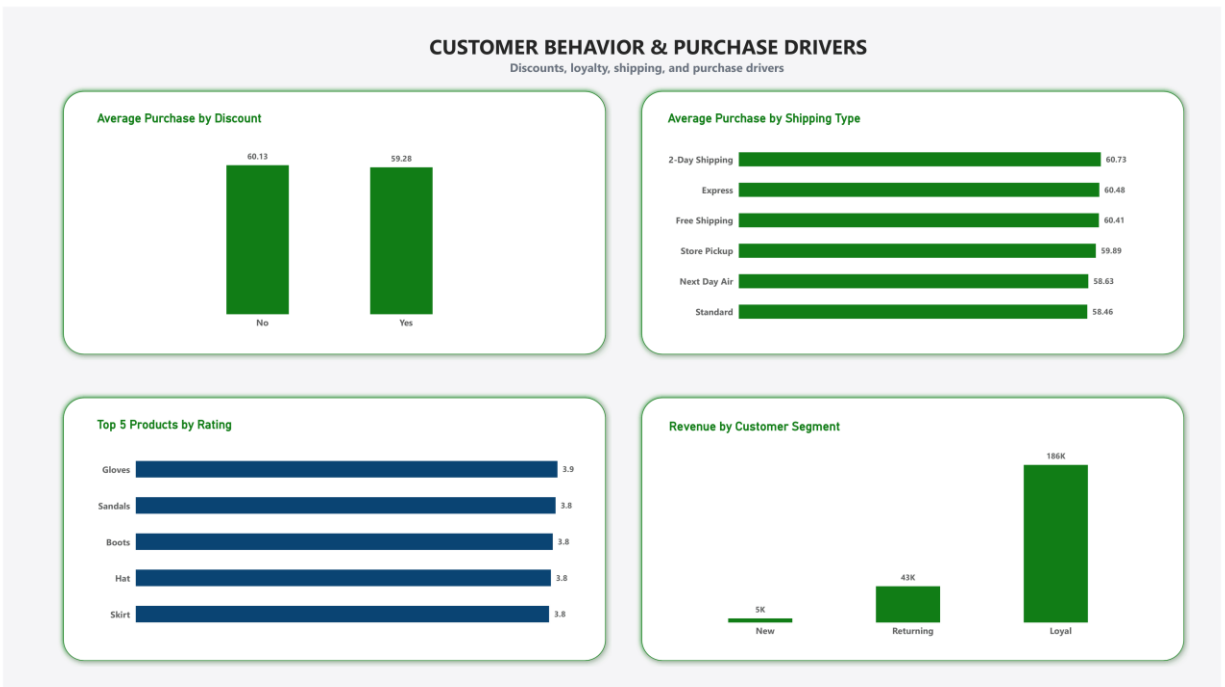
Page 01

Provides an overview of customer volume, revenue, subscription status, demographics, and purchasing behavior.



Page 02

Examines discount behavior, shipping, product ratings, and customer-segment revenue.



04. Findings + Recommendations

Key Findings

Customer base

3.9k records are represented in the analysis, with approximately **27% subscribers** and **73% non-subscribers**.

Revenue

Total revenue is approximately **\$233k**, with different contributions across product categories.

Average purchase

Overall average purchase is **\$59.76**

Customer loyalty

The loyal segment contributes substantially more revenue than new and returning customers in the current analysis.

Product ratings

Gloves, sandals, boots, hat, and skirt are the five products with the highest average ratings in the analysis.

Recommendations

1. Increase subscription engagement

Evaluate targeted benefits and offers for non-subscribers.

2. Strengthen retention

Use targeted strategies to encourage progression from new, returning, and loyal.

3. Review discount strategy

Evaluate discount usage by product and customer behavior to balance promotions with revenue objectives.

4. Promote highly rated products

Use highly rated products in marketing and product-placement strategies.

5. Target customer segments

Use customer behavior and demographics to create more relevant marketing campaigns.

By analyzing who customers are, what they buy, how often they purchase, and which factors influence their spending, the company can identify trends, segment customers, personalize engagement, target marketing, promotions, and optimize its product strategy.